Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	White Cloud
Project / Solution Name	LumiPath
Name (Leader)	Elson Lim Zhong Qin
Email & Phone (Leader)	elsonlimzhongqin.lzq@gmail.com | +60 12-776-1455
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Live Prototype URL	https://lumipath.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/elsonlim2006/LumiPath

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

LumiPath utilizes a lightweight, decoupled web-and-serverless architecture designed for zero-latency career navigation. The frontend is a responsive single-page application built with HTML5, CSS3, and vanilla JavaScript, providing an interactive, zero-scroll user experience for resume uploading, profile calibration, and dual-scenario exploration. The backend is powered by Python and Flask, orchestrating three core decoupled services:
- ProfileParserService extracts text from resumes in-memory with pypdf and structures worker profiles
- MatcherService evaluates quantitative career compatibility against O*NET 28.1 and TalentCorp Critical Occupations List (COL) benchmarks using a weighted formula (0.7 skill overlap + 0.3 tenure factor)
- PathwayService synthesizes chronological 3-phase reskilling roadmaps linked to Malaysia HRD Corp claimable courses. External LLM intelligence (Google Gemini API) is safeguarded by strict whitelisting and an offline heuristic fallback cache, guaranteeing crash-proof execution.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.10+	Core application runtime & backend service logic	PSF License
Flask 3.0.3	Web server & RESTful API routing	BSD-3-Clause
flask-cors 4.0.1	Cross-Origin Resource Sharing middleware	MIT
requests 2.32.3	HTTP client for Gemini API and cloud integration	Apache-2.0
pypdf 4.3.1	In-memory text extraction from uploaded resume PDFs	BSD-3-Clause

Name & Version	Purpose in Project	License
HTML5 / CSS3 / Vanilla JS	Client-side interactive single-page user interface	W3C / MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API (gemini-flash-lite-latest)	Generative career pathway summarization & resume skill extraction	Free Tier

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale Deployments	24/7 public production hosting for web application & REST API	Replit Core / Global
RunPod Serverless	Containerized background execution endpoint for diagnostic workloads	Serverless Tier

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gemini-flash-lite-latest	Google	Pre-trained	Google Terms of Service

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET 28.1 Database	https://www.onetcenter.org	Public	CC BY 4.0	N
Critical Occupations List (COL)	TalentCorp Malaysia (https://www.talentcorp.com.my)	Public	Government Open Data	N
Hays Asia Salary Guide 2025	Hays Asia Industry Report	Public	Open Reference	N
Course & Certification Catalog	NIOSH Malaysia / Coursera / edX	Public	Open Reference	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Curated Sample Profiles	In-house Synthetic Data (src/data/sample_profiles.json)	Synthetic	MIT	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Antigravity (Google DeepMind)	Code scaffolding, modular refactoring, and automated test suite design	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

All AI-generated code and copy underwent rigorous human engineering review. Unit tests (tests/test_utils.py) and end-to-end integration tests (tests/test_e2e.py) were executed to verify mathematical formulas (0.7 skill overlap + 0.3 tenure factor), schema conformance across 7 root contract keys, and zero-hallucination skill whitelisting before merging into production.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Platform credits & Autoscale hosting	Hosted the 24/7 public web prototype (https://lumipath.replit.app) and collaborative workspace.
RunPod	Serverless container execution credits	Configured and deployed runpod/handler.py serverless endpoint for containerized diagnostic workloads.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provided seamless continuous deployment, zero-setup collaborative editing, and persistent public hosting for live judging. RunPod demonstrated decoupled serverless container execution for distributed workloads. While the core Python Flask logic is container-portable and can run on standard VPS, sponsor platforms accelerated zero-latency development, deployment reliability, and isolated sandbox execution.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No personal data is collected or persisted. Uploaded resumes (PDF/TXT/DOCX) are parsed entirely in-memory using stateless regex and lightweight API extractions. No files, personal identifiers, or profile payloads are written to disk or permanent databases. Synthetic benchmark profiles (e.g. Wong Swee Tiam) utilize sanitized fictitious contact details under example.com domains. All external API requests transmit only stripped skill arrays without personal metadata.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Primary technical risks include LLM hallucinations, external API latency, and provider rate limits. Mitigations implemented include: (1) Deterministic Whitelisting: Dynamic trend signals and role recommendations are strictly constrained by authoritative benchmarks (O*NET 28.1, TalentCorp COL). (2) Resilient Dual-Layer Fallback: In cases of API timeout (>4s) or network failure, an offline heuristic cache instantly serves validated roadmap recommendations, guaranteeing 100% crash immunity.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]
10/24/2026